

Cheng (Kevin) Shao

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EDUCATION

University of Waterloo

B.A.Sc. Electrical Engineering, Honours, Co-op

Waterloo, ON

Expected May 2030

TECHNICAL SKILLS

- **Languages/Firmware:** C++, Arduino (Wiring)
- **Hardware & Tools:** Arduino UNO R4, Oximeters, Multimeters, Soldering, Power Tools, Automotive Diagnostics
- **Relevant Skills:** Circuit Assembly, Safety Compliance, Team Collaboration, Technical Documentation

EXPERIENCE

Project Team Member

Remote

Waterloo Experience (WE) Accelerate Program

Winter 2026

- Conducted an accessibility and usability audit of five faculty co-op community pages to identify UI/UX inconsistencies and critical accessibility failures.
- Developed comprehensive content modules for a digital accessibility micro-course, focusing on accessible HTML, CSS, and inclusive web design practices.
- Collaborated within a cross-functional project team to establish manifestos and evaluate digital environments for strict AODA compliance and adherence to WCAG standards.

Manual Labour Worker

Nanaimo, BC

Vanderkooi Farm

Dec 2024 – Jul 2025

- Collaborated with a fast-paced team (4-6 individuals) to carry out large-scale chicken transfer operations.
- Processed more than 20,000 chickens each time while keeping to strict safety and humane handling requirements.
- Maintained high performance during intense seasonal peak periods, demonstrating resilience and reliability.

Room Maintenance Worker

Nanaimo, BC

Mapleview B&B

Summers 2022–2025

- Managed guest accommodations and property arrangements for arriving visitors while keeping a high degree of organization and cleanliness.
- Handled listing coordination and guest communication, addressing concerns quickly to achieve positive reviews.

PROJECTS

Smartwatch Sleep Monitor

Sep 2025 – Dec 2025

ECE 198 Design Project, Arduino, C++

- Prototyped a wearable sleep tracking device using an Arduino UNO R4 WiFi and oximeter sensor to address hospital-induced delirium.
- Programmed firmware to detect sleep cycles by analyzing real-time heart rate variability against specific BPM thresholds (Start < 55 BPM, End > 65 BPM).
- Implemented IoT functionality to transmit automated daily sleep reports and data arrays directly to caregivers via Gmail/Wi-Fi integration.
- Validated system reliability through component testing, confirming accurate heart rate logging, state switching, and successful email data delivery.

Trades & Mechanical Projects

Sep 2024 – Feb 2025

VIU Trades Program

- **Electrical:** Fabricated electrical extension units from scratch, interpreting technical schematics and ensuring compliance with CSA grounding and safety standards.
- **Mechanical:** Performed automotive maintenance, including brake pad replacements and oil/filter changes, gaining practical understanding of vehicle systems and workshop safety.
- **Woodworking:** Fabricated custom products using industrial power tools, applying precision finishing techniques and analyzing material properties.
- **Welding:** Utilized thermal cutting and welding techniques to separate and fuse metal components, focusing on heat control and structural integrity.